

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

B.Tech (Electronics and Telecommunication Engineering) SEMESTER - 5 Winter 2025 (Regular)

Course : Bachelor of Technology (Electronics and Telecommunication Engineering) Branch : Engineering and Technology

Semester : SEMESTER - 5

Subject Code & Name: BTETOE505B - ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6
4. Use of non-programmable scientific calculators is allowed.
5. Assume suitable data wherever necessary and mention it clearly.

Q1. Objective type questions. (Compulsory Question)

12

- 1 Which approach in AI focuses on thinking like humans?
a) Cognitive modeling b) Rational agent c) Laws of thought d) Neural networks
- 2 The structure of an agent includes
a) Sensors and actuators b) Transistor c) Database d) Clock
- 3 Uninformed search does not use
a) Heuristic b) Queue c) Stack d) Path cost
- 4 Breadth-first search is guaranteed to find
a) Optimal solution b) Any solution c) Approximate solution d) Fastest path
- 5 Greedy best-first search chooses nodes based on
a) Heuristic b) Path cost c) Depth d) Random
- 6 Bidirectional search starts search from
a) Both initial and goal b) Initial only c) Goal only d) Random node
- 7 Minimax algorithm is used for
a) Optimal decision making b) Heuristic evaluation
c) Forward chaining d) Planning
- 8 Forward pruning in games involves
a) Ignoring unlikely moves b) Increasing search depth
c) Using neural nets d) Random selection
- 9 Forward chaining is a method of
a) Inference b) Learning c) Searching d) Planning
- 10 Backtracking search in CSPs is used to
a) Find solutions b) Generate heuristics c) Learn patterns d) Prune nodes
- 11 Decision trees are used for
a) Classification b) Regression only c) Neural networks d) Heuristic search
- 12 Ensemble learning improves performance by
a) Combining multiple models b) Using single model
c) Increasing depth d) Reducing training

- Q2. Solve the following.
- A) Explain the difference between thinking humanly, thinking rationally, and acting rationally in AI. 6
 - B) Discuss the concept of rational agents and the role of environments in AI. 6
- Q3. Solve the following.
- A) Explain the role of problem-solving agents in AI. Differentiate between well-defined problems and real-world problems, with examples of problem formulation. 6
 - B) What are heuristic functions? Discuss the properties of a good heuristic and the effect of heuristic accuracy on search performance. 6
- Q4. Solve Any Two of the following.
- A) Explain the Minimax algorithm for optimal decisions in two-player zero-sum games, with a suitable example. 6
 - B) Explain cutting off search, forward pruning, and evaluation functions techniques used in practical game programs. 6
 - C) Explain evaluation functions in games and Kriegspiel as a partially observable game. 6
- Q5. Solve Any Two of the following.
- A) Define Constraint Satisfaction Problems (CSPs). Explain constraint propagation techniques (e.g., arc consistency) and backtracking search for solving CSPs. 6
 - B) Discuss propositional logic, theorem proving, and model checking in knowledge-based agents. 6
 - C) Explain planning as state-space search and the use of planning graphs. 6
- Q6. Solve Any Two of the following.
- A) Explain supervised, online, and ensemble learning. Include examples. 6
 - B) Discuss decision trees, regression, and classification using linear models. 6
 - C) Explain learning with hidden variables (EM Algorithm) and inductive logic programming. 6

*** End ***